EIGSEP Memo 006

Marjum Pass 2026-07: RFI, the two combs, and flagging

EIGSEP Team

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Abstract

What contaminates the Marjum spectra, how to remove it, and which flag product to use. The two combs in the data are named and dated (memo 001 identifies them; this memo places them in time and gives the masks), and the archive labels that had them the wrong way round are corrected. The campaign's default flags are `flags@v3`, a smooth-background detector run on both antennas and on calibration rows; this memo gives its method, coverage and occupancy, and three validations, and records why the earlier products are not the default. Every figure, table and quoted number is generated by `make_products.py` in this directory (Appendix A).

1 Summary

- **Two combs.** The *transmitter* comb (channels $\equiv 0 \bmod 8$, 1.953125 MHz), first seen at 15:36:22 UTC on 07-17 on both antennas, on in 22 episodes (87% of its era). The *box-air self-EMI* comb (1.000 MHz), 07-16 01:18–16:51, box-air only. New masks: `boxair-emi-1mhz` (default) and `tx-comb-teeth`.
- **The box-air self-EMI recurs** for one to two minutes at the start of two calibration visits on 07-17 (08:53, 14:08), with the same 1.000 MHz spacing and again on box-air only (§2.1). It is not the VNA and not on the transmitter's grid.
- **Archive labels corrected**: every product, mask, mode and code path that called the transmitter a "self-comb" or the self-EMI "the TX" now uses the names of Table 3.
- **Flags**: `flags@v3` is the default: 3,395 files (phases B and C), both antennas and calibration rows, no failed files. It excludes 59% of box-air sky cells and 52% of box-gnd's, most of it in the 07-16 self-EMI, the transmitter era and 07-15 (67% and 59% with the advisory high-scatter bit). It recovers half of planted lines at about 6 times the measured scatter, and excludes 93–96% of what an independent detector flags (98–100% with the conservative mask; §5).

2 The two combs

Memo 001 (§Combs) establishes which comb is which; this section records where each is and how to remove it.

The box-air self-EMI has teeth at integer MHz, about 10 dB, seen by box-air only, from 07-16 01:16:57 to 16:49:03 (the first and last affected files close at 01:18:00 and 16:51:10); it vanished when the Panda was power-cycled (memo 001). 436 box-air files carry it; 3 of them fall outside that window (20260717 085537, 20260717 140909, 20260717 141118), and the two on 07-17 are the calibration visits of §2.1. The new mask `boxair-emi-1mhz` removes the window from box-air (inputs 4 and its multiplexer copy 5) by default (Table 1). It replaces `comb-rfi-1p25mhz`, which covered only 01:00–01:30 and was named for a 1.25 MHz comb that does not exist (memo 001).

The transmitter is the wanted signal for beam mapping (memo 005) and contamination for sky work. Box-gnd sees it without antenna modulation, so it times it: 22 on episodes from 15:36:22 on 07-17, on 87% of the time between the first switch-on and the last switch-off (Fig. 1b; on when the 8-channel lock is ≥ 30 with phase 0, per 8-integration block). The mask `tx-comb-teeth` drops whole files in the transmitter era; sky work

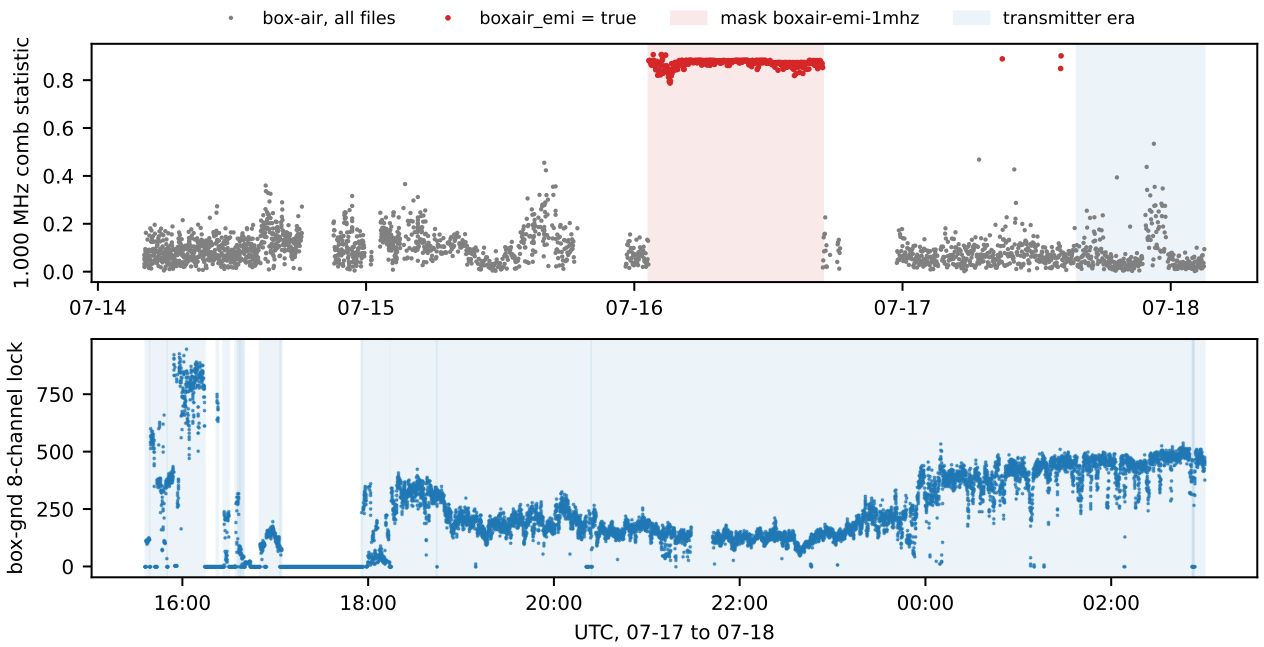


Figure 1: (a) Box-air's 1.000 MHz comb statistic per file (`curation/boxair_emi_presence.jsonl`), files judged to carry the comb in red, with the `boxair-emi-1mhz` mask window and the transmitter era. (b) The transmitter seen by box-gnd, per 8-integration block (`flags/diag/transmitter_per_integration_boxgnd.jsonl`); shading marks its on episodes (`curation/transmitter_transitions_boxgnd.jsonl`).

that keeps those files should instead exclude channels $\equiv 0 \bmod 8$. The per-file mode-table column `transmitter` (on/off/partial) comes from the same box-gnd record.

2.1 The self-EMI in calibration visits on 07-17

Three calibration blocks on 07-17 carry strong narrow lines that look like a comb. Their spacing decides what they are, so it is measured directly: for each integration, the spectrum is divided by its 31-channel running median and the residual is Fourier-transformed across channels; a 1.000 MHz comb puts power at 1/4.096 cycles per channel, the transmitter's grid at 1/8 (and 1/4). Table 2 and Fig. 2 give the result. In the noise-source-on block at 08:53 and in both the noise-source-on and ambient-load blocks at 14:08–14:10, box-air shows a 1.000 MHz comb up to 6,412 times the residual's median power, with nothing at the transmitter's spacing (at most 31) and nothing on box-gnd (at most 4). The 07-16 self-EMI gives the same signature (box-air 4,322 at 1.000 MHz, 10 at 8 channels; box-gnd 4), and the transmitter the opposite (box-air 12 and 773; box-gnd 2,285 at 8 channels). The comb is absent from the VNA sweeps that follow each block and from the sky data on either side. **These are box-air's 1.000 MHz self-EMI, briefly back**, not the transmitter and not the VNA. The identification rests on spacing and on which antenna sees it; what switches the self-EMI on at the start of a calibration visit is not known.

3 The labels that had the combs backwards

Before memo 001 the archive called the transmitter a "digital self-comb" and the self-EMI "the TX comb". Every live name is corrected (Table 3); old products are removed and their generators renamed, and each regenerated product records `renamed_from` in its provenance. Names whose meaning inverts get new names rather than aliases: selecting `tx-comb` now stops with an error naming its two successors. Historical notebooks and their HTML exports keep the old names; their directory READMEs point here.

Table 1: The comb masks in `eigsep_data.select_files` (first sentence of each reason).

mask	window (file close, UTC)	inputs	default	reason
`boxair-emi-1mhz`	07-16 01:18–07-16 16:51	4, 5	yes	box-air's own 1.000 MHz comb EMI (teeth at integer MHz, ∼10 dB, walks across the channel grid; box-air only, 07-16 01:16:57-16:49:03, gone after the Panda power cycle).
`tx-comb-teeth`	07-17 15:37–07-18 03:24	all	no	Beam-mapping transmitter comb: teeth at 1.953125 MHz = exactly 8 channels (channels = 0 mod 8; arm 0 on channels = 0 mod 16, arm 1 on = 8 mod 16), on BOTH antennas from 07-17 15:36 to the end of the data (memo 001).

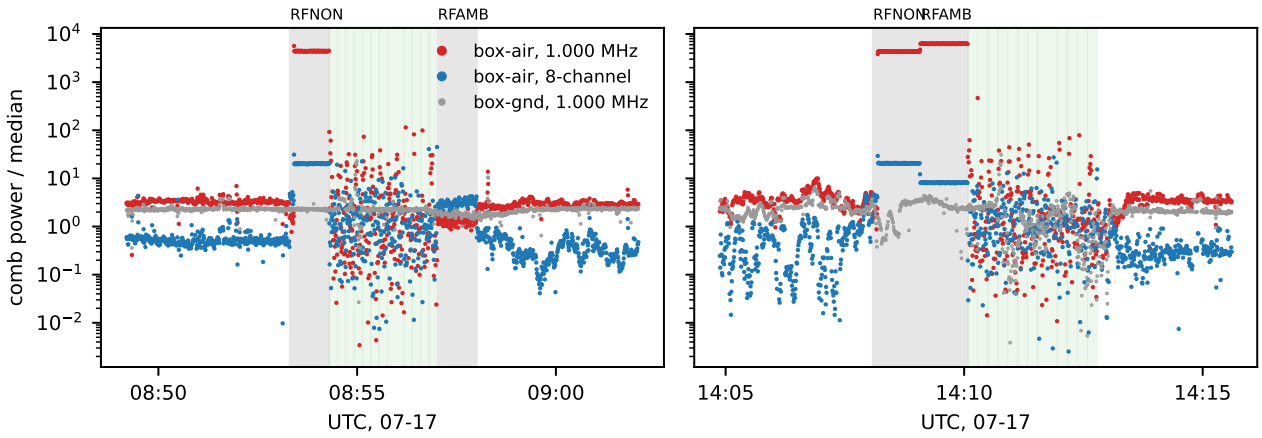


Figure 2: Comb power per integration through the two 07-17 calibration visits: box-air at the 1.000 MHz and 8-channel spacings, box-gnd at 1.000 MHz. Grey: noise-source-on (RFNON) and ambient-load (RFAMB) blocks; green: VNA states.

4 Flag products

The campaign default is `flags@v3` (§5). Three older products remain on disk for provenance.

`flags@v0` (`flags/v0/`; all inputs) flags 4.98% of sky samples with a per-channel temporal-median detector and a comb matched filter. Its bit 1 (`tx_comb`) is never set: on 07-17 and 07-18 the transmitter's teeth are in bit 2. Its temporal-median detector cannot see a line that is on all the time.

`flags@v2` adds bit 8 (`dpss_residual_outlier`) to v0. Its threshold is not centred, so it flags about 27% of pixels where a centred one flags about 5% (flags/v2 README); it covers about 40% of file–input pairs; and it runs no detector on calibration rows.

`flags@v3-beta.1` introduced the method of §5 on box-air only, and had two faults. 687 of its 3,395 files failed to fit and are unprocessed, and its main RFI bit (bit 2, `positive_auto_excess`) set 13–39% of sky cells per day, because it thresholds $z = (d/m - 1)\sqrt{2\,\Delta t\,\Delta\nu}$, the residual in units of radiometer noise, at 6: a 1.2% excess. The robust scatter of z on the cells it keeps is 2.2 on average (10th–90th percentile 1.4–3.5; Fig. 3), not 1, and those cells have already been cut at $z > 6$, so the smooth model's own error was flagged as RFI.

Table 2: Comb power (peak over rows, in units of the residual's median spectral power) in the 07-17 calibration blocks, and the rows where box-air's 1.000 MHz power exceeds 50.

block (UTC)	state	rows with 1 MHz comb	box-air 1 MHz	box-air 8-ch	box-gnd 1 MHz
08:53:18–08:54:17	RFNON	99 / 110	5,651	31	3
08:57:01–08:58:00	RFAMB	0 / 110	2	4	2
14:08:05–14:09:03	RFNON	99 / 110	4,409	29	4
14:09:05–14:10:04	RFAMB	110 / 110	6,412	8	4
all VNA states		10 / 567	115	45	31
RFANT either side		0 / 1582	14	5	11

Table 3: Corrected names. Products are under `marjum-2026-07/`; scripts under `data-analysis/scripts/marjum-2026-07/`.

old	new
mode-table column `tx_comb` (it tracked the 1 MHz self-EMI)	`boxair_emi`; and a new column `transmitter`
`select_files` mode `tx-comb`, CLI `-tx-comb`	`boxair-emi` / `transmitter`; `tx-comb` is an error
mask `digital-self-comb`	`tx-comb-teeth` (old name a deprecated alias)
mask `comb-rfi-1p25mhz` (01:00–01:30)	`boxair-emi-1mhz` (01:18–16:51, box-air, default)
`curation/tx_presence.jsonl` (`tx_on`)	`curation/boxair_emi_presence.jsonl` (`boxair_emi`)
`curation/self_comb_presence.jsonl`	`curation/transmitter_presence.jsonl`
`curation/tx_state_matched.jsonl`	`curation/boxair_emi_matched.jsonl`
`curation/tx_comb_eras.jsonl`	`curation/comb_eras.jsonl`
`curation/self_comb_transitions_boxgnd.jsonl` (unmerged)	`curation/transmitter_transitions_boxgnd.jsonl` (now tracked)
`flags/diag/self_comb_per_integration_boxgnd.jsonl`	`flags/diag/transmitter_per_integration_boxgnd.jsonl`
`identified_as: "digital self 1.953125 MHz"`	`"transmitter 1.953125 MHz"`
flagging detectors `digital_self`, `panda_emi`	`transmitter`, `boxair_emi`
scripts `scan_tx_presence`, `build_self_comb_presence`, `scan_tx_comb`, `build_tx_state_matched`, `*_self_comb_*`	`scan_boxair_emi_presence`, `build_transmitter_presence`, `scan_comb_eras`, `build_boxair_emi_matched`, `*_transmitter_*`

5 `flags@v3`

Method (`eigsep_data.rfi_supported`, revision `supported-dpss-v3-beta-lowband-5-scatter-advisory`; `eigsep_data/docs/rfi_v3_beta.md`). For each flagged antenna, a smooth background — DPSS modes 50 ns wide in frequency times modes 1 mHz wide in time, fitted robustly in each contiguous segment — is subtracted, and a cell is flagged if its residual is a positive outlier (bit 2), if the cross-correlation with the other antenna changes against its own smooth background (bit 4), if a report band or 8-channel tile, or a clock-aligned group of channels sharing a residue mod 8, 4 or 2, rises together (bits 5, 6). Bit 7 marks cells the background cannot support, and bit 9 is advisory (below). The changes from `v3-beta.1`:

- bit 2 divides z by each channel's measured scatter (never below 1) before the threshold of 6;
- a channel whose untruncated residual scatter exceeds 5 times radiometer noise gets the advisory bit 9 (`high_scatter`): the background cannot describe it and detection there is weak, but the cell is not excluded by default;
- calibration rows (RFNON, RFAMB) are flagged against their own backgrounds, keeping bit 0;
- both antennas are flagged: box-gnd with box-air as its reference;
- files recorded with the switch daemon off, when the switch sat on the antenna, are flagged as sky (1,280 files);
- a cross-correlation channel that does not converge loses only the cross detector, uneven or repeated timestamps get a usable time basis, and the CG budget is 2000 iterations at the same tolerance: no file fails.

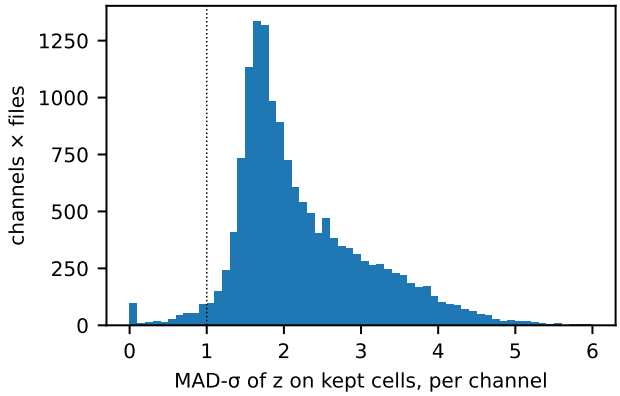


Figure 3: Per-channel robust scatter of the radiometer-normalized residual on the cells `flags@v3-beta.1` keeps, on 26 sample files. Radiometer noise alone gives 1.

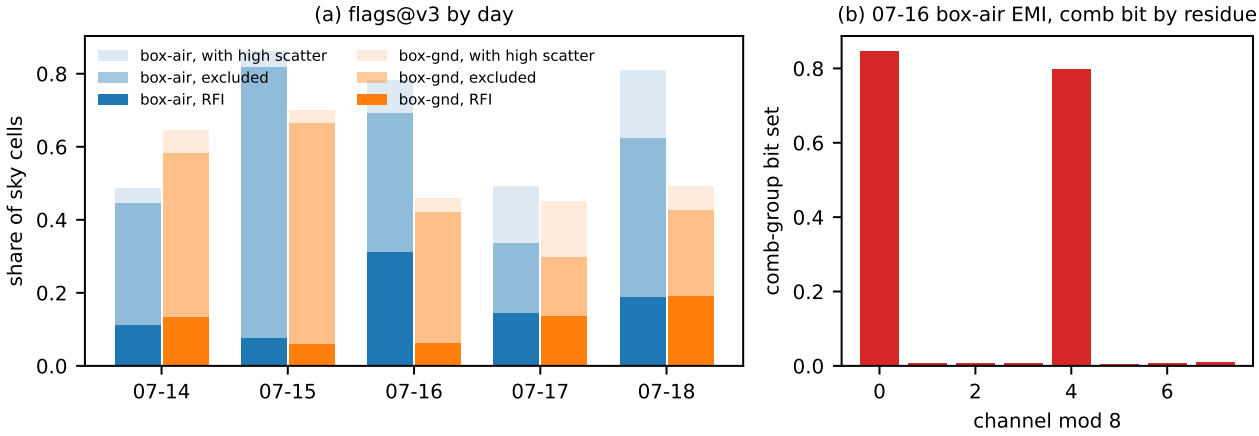


Figure 4: (a) Share of sky cells flagged as RFI (dark), excluded by the default mask (medium) and by the conservative mask including high scatter (light) in `flags@v3`, per day and antenna. (b) Share of box-air cells with the comb-group bit during the 07-16 self-EMI, by channel residue mod 8.

Coverage. 3,395 files, 3,395 of them for both antennas: every file the antenna-resolution policy admits (phases B and C). Phase A is excluded until its antenna-to-input mapping is known (memo 001).

Occupancy. Over the campaign the default mask excludes 59% of box-air sky cells (15% as RFI) and 52% of box-gnd's (10% as RFI); adding the advisory high-scatter bit, the conservative mask excludes 67% and 59% (Table 4, Fig. 4). Exclusion is high in three places, for different reasons: the 07-16 self-EMI (real RFI, flagged, and high scatter where the walking comb passes); the transmitter era of 07-17/18, where broadband variations the smooth background cannot describe raise the scatter on both antennas, including the rotating raster; and 07-15, where many cells fail the background's support tests and the band carries dense narrow lines (113–128, 144–156 MHz), much stronger on the elevated box-air than on box-gnd. The comb detector fires on channel residues 0 and 4 during the 07-16 self-EMI (84% and 80% of cells; at most 0.9% elsewhere), the 1.000 MHz comb's aliases on the 4.096-channel spacing: there is no 8-channel comb on 07-16.

Validation. Three tests, each able to fail:

- **Planted lines.** Narrow lines planted in sky cells neither setting excluded, on four ten-file batches and both antennas (14,400 lines; Fig. 5, Table 5): outside the raster, half are recovered at 13 times radiometer noise, which is about 6 times the measured scatter, and all at 40. The `v3-beta.1` settings recover half at

Table 4: Shares of sky cells in `flags@v3`: excluded by the default mask (any bit but 9), flagged as RFI (bits 2, 4, 5, 6), unsupported (bit 7), with the advisory high-scatter bit (9), and excluded by the conservative mask (any bit); and the share of calibration-row cells flagged as RFI, outside the two self-EMI visits. Bits overlap.

day	antenna	excluded	RFI	unsupported	high scatter	conservative	cal. RFI
07-14	box-air	44%	11%	29%	14%	49%	6.6%
07-14	box-gnd	58%	13%	41%	19%	65%	2.8%
07-15	box-air	82%	7%	58%	15%	86%	–
07-15	box-gnd	67%	6%	30%	9%	70%	–
07-16	box-air	69%	31%	36%	37%	78%	31.8%
07-16	box-gnd	42%	6%	33%	9%	46%	4.9%
07-17	box-air	33%	14%	15%	34%	49%	0.4%
07-17	box-gnd	30%	13%	12%	31%	45%	5.9%
07-18	box-air	62%	19%	39%	53%	81%	0.3%
07-18	box-gnd	42%	19%	18%	24%	49%	14.7%

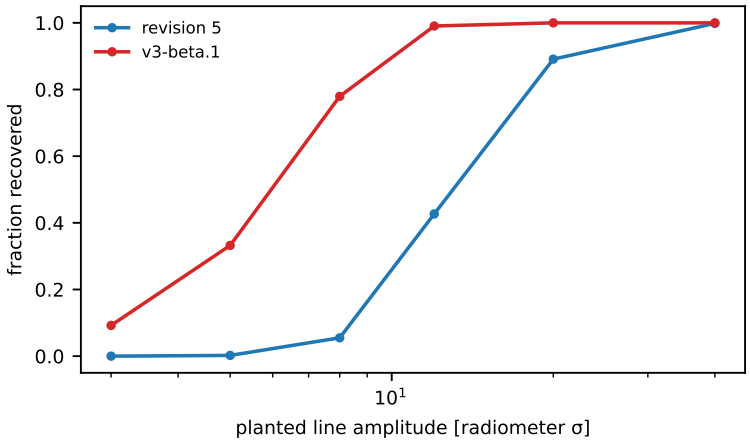


Figure 5: Fraction of planted narrow lines flagged as RFI against their amplitude in radiometer noise, averaged over the non-raster sample batches and both antennas, for the product's settings and those of `flags@v3-beta.1`.

6, only because they also flag the scatter. In the raster, where the scatter is 8–11 times radiometer noise, lines at 20 times radiometer noise are recovered 0% (box-air) and 16% (box-gnd) of the time: that is what bit 9 marks.

- **An independent detector.** Of the box-air cells that `flags@v0`'s targeted detectors flag (FM scatter, airplane, ORBCOMM), `flags@v3`'s default mask excludes 93–96% on every day, and its conservative mask (with bit 9) 98–100% (Table 6): what the default mask misses lies mostly in high-scatter channels, where detection is weak. 235 files containing empty rows are left out: there `v0`'s temporal median includes the zeros and it flags ordinary rows (in `corr_20260715_115658Z`, 134 of 140 non-empty rows as "airplane", with no excess at 73–88 MHz). Most of what the product flags beyond `v0` is concentrated in a few channels flagged most of the time: persistent lines, which `v0`'s temporal-median detector cannot see.
- **A null.** Calibration rows see a load or a noise source, not the sky, so RFI flags there come from the detector or from interference inside the receiver (Table 4, last column, which leaves out the two 07-17 visits that carry the self-EMI). Box-air's are 0.3–0.4% on 07-17 and 07-18; 32% on 07-16, where box-air's own 1.000 MHz self-EMI is in its calibration states too; and about 7% on 07-14, not explained. Box-gnd's are 3–15%, highest in the transmitter era, which box-gnd picks up by conduction even on its loads. Where nothing is conducted the detector's own false-positive rate is below 1%.

Table 5: Planted-line recovery per sample batch and antenna, and the shares of sky cells flagged as RFI and excluded, for the product's settings and those of `flags@v3-beta.1`.

batch, antenna	3σ	5σ	8σ	12σ	20σ	40σ	RFI	excluded
07-16 box-air EMI era, box-air, revision 5	0.00	0.00	0.02	0.25	0.68	0.99	48%	57%
07-16 box-air EMI era, box-air, v3-beta.1	0.10	0.36	0.76	0.98	1.00	1.00	52%	63%
07-16 box-air EMI era, box-gnd, revision 5	0.00	0.00	0.05	0.39	0.91	1.00	5%	13%
07-16 box-air EMI era, box-gnd, v3-beta.1	0.09	0.35	0.75	1.00	1.00	1.00	8%	19%
07-17 night, box-air, revision 5	0.00	0.00	0.05	0.40	0.91	1.00	13%	23%
07-17 night, box-air, v3-beta.1	0.07	0.30	0.80	0.99	1.00	1.00	18%	31%
07-17 night, box-gnd, revision 5	0.00	0.01	0.10	0.54	0.93	1.00	9%	20%
07-17 night, box-gnd, v3-beta.1	0.08	0.35	0.80	0.99	1.00	1.00	15%	27%
07-17 day, box-air, revision 5	0.00	0.00	0.03	0.37	0.94	1.00	10%	19%
07-17 day, box-air, v3-beta.1	0.13	0.33	0.76	0.99	1.00	1.00	14%	25%
07-17 day, box-gnd, revision 5	0.00	0.00	0.07	0.61	0.98	1.00	8%	18%
07-17 day, box-gnd, v3-beta.1	0.09	0.30	0.81	0.99	1.00	1.00	10%	20%
07-17 transmitter raster, box-air, revision 5	0.00	0.00	0.00	0.00	0.00	0.71	26%	71%
07-17 transmitter raster, box-air, v3-beta.1	0.18	0.35	0.61	0.97	1.00	1.00	46%	92%
07-17 transmitter raster, box-gnd, revision 5	0.00	0.00	0.00	0.01	0.16	0.93	24%	56%
07-17 transmitter raster, box-gnd, v3-beta.1	0.12	0.25	0.65	0.96	1.00	1.00	45%	79%

Table 6: `flags@v0`'s targeted detections against `flags@v3`, box-air sky cells, outside the 235 files with empty rows: the share excluded by the default and conservative masks, and the share of the product's RFI cells that `v0` does not flag, with the fraction of those in the 100 most-flagged channels.

day	v0 targeted cells excluded		product RFI	of which in
	default	conservative	not in v0	100 channels
07-14	95.8%	99.2%	89%	49%
07-15	95.5%	99.5%	85%	28%
07-16	92.7%	98.7%	51%	44%
07-17	93.1%	98.1%	72%	53%
07-18	96.3%	98.7%	41%	48%

6 What to use

- **File selection**: `eigsep_data.select_files` with its defaults, which include `boxair-emi-1mhz`.
- **Per-sample flags**: `flags@v3`, through `load_bundle(..., products=["flags@v3"])`; exclude every cell with any bit set except bit 9 (`RFIResult.mask`). Add bit 9 for a conservative mask, and always where the antenna moves (the raster, scans), where detection is weak. Calibration rows carry bit 0 plus their own RFI bits. **Do not mask with the RFI bits alone** (2, 4, 5, 6): non-positive samples, including int32 accumulator wraps of about -2×10^9 counts on strong lines and empty rows, carry only bit 1 (`invalid_input_or_domain`), and left in they pull a weighted mean below zero (memo 004).
- **Beam mapping** keeps its own raster flags (memo 005): in the raster the product's detection is weak (bit 9 nearly everywhere).
- **Transmitter era**: exclude channels $\equiv 0 \bmod 8$ for sky work even where cells are kept.

7 Open questions

- What switches box-air's 1.000 MHz self-EMI on, on 07-16 and at the start of two calibration visits on 07-17. A bench test of the Panda and the box-air electronics would identify it physically.
- The source of the dense narrow lines that dominate 07-15 and the persistent lines at 150–162 MHz: external transmitters, by their bands and their strength on the elevated antenna, but not identified.

- A background that follows a moving antenna, so the raster and other scans can be flagged rather than excluded. A smooth one in elevation does not do it: harmonics of elevation times the slow time modes, fitted exactly on the two 07-17 raster batches (`eigsep_data` branch `rfi-pointing-line`, `75c2eb6`; `validate_pointing_bg.py`), adds at most 0.004% new flags as a line detector, because `v3` already excludes 71–76% of raster sky cells; used instead of `v3`'s background there it would exclude only 26–36%, but in the cells it frees planted lines at 30 and 100 times radiometer noise are recovered 4–8% and 25–36% of the time (90th-percentile channel scatter 330–548 times radiometer noise). The beam model (memo 005) is the likelier background.
- Phase A, once its antenna-to-input mapping is known.

Revision history

2026-10-03 First version. Replaces legacy MEMO-007 (comb differencing) and MEMO-009 (flag masks v0). Adds the corrected comb names and masks, the calibration-visit self-EMI, the evaluation of `flags@v3-beta.1`, and the revision-2 point threshold on a sample.

2026-10-04 `flags@v3` (v3-beta revision 5: both antennas, calibration rows, daemon-off files as sky, an advisory bit where the background cannot describe a channel, no failed files) is the campaign default. Replaces the revision-2 sample comparison with the product's coverage, occupancy and validation.

2026-10-05 The default product is released as `flags@v3`: the same payloads as `v3-beta.3`, renamed; later improvements will be new versions.

A Reproducing this memo

`python validate_v3_point.py` (slow) writes `validate_v3_point.npz`, and `python validate_pointing_bg.py`, with `eigsep_data` branch `rfi-pointing-line` on the path, writes `validate_pointing_bg.json`; then `python make_products.py` writes every figure, table and number, and `pdflatex memo-006.tex` twice. Set `EIGSEP_CAMPAIGN_ROOT` to the campaign directory that holds the raw data and gitignored products when running from a worktree. The product itself was made by `eigsep-rfi-v3-beta` (`eigsep_data.rfi_run`) with `-all -files-per-batch 10`, once with `-air-antenna box-air` and once with `-air-antenna box-gnd -ground-antenna box-air`; its manifest records the parameters, source hash and resolution policy of every file. The gitignored files read are listed with their SHA-256 in `tables/inputs.json`.

gitignored input	SHA-256 (first 16)
`data/` (39 files; digest of their SHA-256s)	`9886394afed84014`
`derived/smooth_model/v3-beta.1/` (26 files; digest of their SHA-256s)	`c45a2d97a4016c73`
`flags/v0/` (5 files; digest of their SHA-256s)	`15057ca2e553de61`
`flags/v3/` (6 files; digest of their SHA-256s)	`f416815da13e2a93`
`flags/v3-beta.1/` (5 files; digest of their SHA-256s)	`46254c1f2e68aec6`
`validate_v3_point.npz`	`fc7dd641af2d6e32`
`validate_pointing_bg.json`	`a7f7a0e6ba6eb473`
eigsep commit	`7864eaf`
eigsep_data commit	`e941c51`
generated	2026-10-06 15:05 UTC